2014 F=ma Exam: Problem 6

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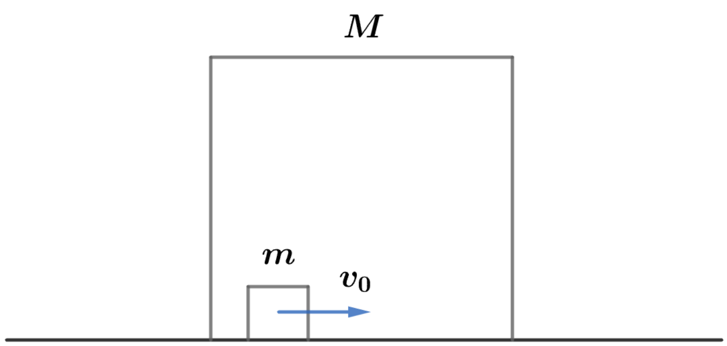

We will approximate the block's position x with the position x_{CM} of the CM. Since both the block and the CM is within the cube of mass M, we know

$$|x - x_{\text{CM}}| \leq 5\,\text{m}$$

To find x_{CM}, note that there is no net external force on the cube-block system so linear momentum is conserved. Since $p_{\text{tot}} = M_{\text{tot}} v_{\text{CM}}$, we know v_{CM} is constant. We have

$$v_{\text{CM}} = \frac{m v_0}{m + M} = \frac{(2\,\text{kg})(5\,\text{m/s})}{2\,\text{kg} + 10\,\text{kg}} = \frac{5}{6}\,\text{m/s}$$

so at $t = 60\,\text{s}$,

$$x_{\text{CM}} = v_{\text{CM}} t = \left(\frac{5}{6}\,\text{m/s}\right)(60\,\text{s}) = 50\,\text{m}$$

Thus, the answer is $\boxed{\text{B}}$.